Introduction to MongoDB

- MongoDB is a NoSQL database that stores data in a flexible, JSON-like format called BSON (Binary JSON).
- It is document-oriented, meaning data is stored as documents in collections instead of rows and tables.
- MongoDB supports dynamic schemas, allowing fields to vary between documents in the same collection.
- MongoDB provides features like indexing, aggregation pipelines, and replication for high performance and reliability.
- Commonly used for applications requiring flexible data models

MongoDB Documents Example

```
_id: ObjectId('675013fb3bfb9415a88f52a9')
name : "Person"
email : "person@gmail.com"
createdAt : 2024-12-04T08:34:03.781+00:00
__v : 0
age : 40
```

```
_id: ObjectId('67501449e9e2a8526f4edc9f')
name : "Thapa"
email : "thapa@gmail.com"
createdAt : 2024-12-04T08:35:21.569+00:00
__v : 0
```

Key Concepts of MongoDB

- **Collections**
  - A collection is a group of documents in MongoDB, similar to a table in relational databases.
  - Collections do not enforce a fixed schema, allowing documents to have different fields.
  - Example: A "users" collection can store data about different users.
- **Documents**
  - A document is the basic unit of data in MongoDB, stored in a JSON-like format.
  - Documents contain key-value pairs, where each key is a field, and the value is the data.
  - ```
    {
      "name": "John Doe",
      "email": "john@example.com",
      "age": 25
    }
    ```

Key Concepts of MongoDB

- **BSON (Binary JSON)**
  - BSON is the binary-encoded format MongoDB uses to store data.
  - It extends JSON by adding data types like Date, Binary, and others.
  - BSON is efficient for data storage and supports faster parsing compared to JSON.
  - Example: A BSON document might store a Date field that is not natively supported in JSON.

- This is not a MongoDB course. Hence, we are just learning through surface essential for now. You can learn in detail from our course.
    ```

Installing and Setting Up MongoDB

- **Download MongoDB**
  - You can download **MongoDB Community** version from this link: https://www.mongodb.com/docs/manual/administration/install-community/
  - Choose the version compatible with your operating system (Windows, macOS, Linux).
- Install MongoDB
  - Follow the installer's instructions to install MongoDB on your system.
  - MongoDB Compass is also installed by default. If it's not installed, then you can install manually by searching online.
  - MongoDB Compass is just a software for interacting with your MongoDB database.
  - Mongosh is a command-line tool for interacting with MongoDB programmatically.
    - This is not installed by default, but you can install yourself if you want.

MongoDB Drivers

- MongoDB drivers are language-specific libraries that allow applications to interact with MongoDB databases.
- They provide APIs for connecting, querying, and managing data within MongoDB.
- MongoDB drivers are available for various programming languages, including Node.js, Python, Java, C++, and more.
- The official MongoDB Node.js driver is mongodb, used for connecting to and performing CRUD operations in Node.js.
- https://www.mongodb.com/docs/drivers/
- As we are using Node.js, we will install **mongodb** npm package.